

CYCLONES, STORM SURGE AND COASTAL PREPAREDNESS

COMPOUND CYCLONE HAZARD MAP → CYCLOGENESIS BOUNDARY → COASTAL WATER FIREWALL → IMD INFORMATION RAIL → COLOUR-STAGE MATRIX → WARNING-TO-EVACUATION CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 COMPOUND CYCLONE HAZARD MAP

COMPOUND CYCLONE HAZARD MAP STRUCTURE DEBRIS POWER TELECOM RIVERINE FLASH PLUVIAL URBAN FLOOD STORM SURGE + WAVES COASTAL INUNDATION EROSION TRANSPORT HEALTH WATER AND LIVELIHOODS

WIND → structure debris power telecom

RAIN → riverine flash pluvial urban flood

STORM SURGE + WAVES → coastal inundation erosion

CASCADE → transport health water and livelihoods

HAZARD / RISK / RESILIENCE CONCEPT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

INSTITUTIONAL OR COMMUNITY RESPONSE

- STORM SURGE + WAVES → coastal inundation erosion
- CASCADE → transport health water and livelihoods

CAPACITY / STATUS / EVIDENCE LIMIT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE + WAVES → coastal inundation erosion.

01

STAGE 01 CYCLOGENESIS BOUNDARY

CYCLOGENESIS BOUNDARY WARM WATER + INSTABILITY + MOISTURE CORIOLIS + DISTURBANCE + LIMITED SHEAR CYCLONIC SYSTEM DM FOCUS EXPOSURE WARNING ACTION CONTINUITY

WARM WATER + INSTABILITY + MOISTURE

CORIOLIS + DISTURBANCE + LIMITED SHEAR

CYCLONIC SYSTEM

DM FOCUS → exposure warning action continuity

HAZARD / ACTOR / INSTRUMENT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

DECISIVE RISK OR MANDATE DISTINCTION

- CYCLONIC SYSTEM
- DM FOCUS → exposure warning action continuity

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CYCLONIC SYSTEM; DM FOCUS → exposure warning action continuity.

02

STAGE 02 COASTAL WATER FIREWALL

COASTAL WATER FIREWALL STORM SURGE CYCLONE WIND AND PRESSURE ASTRONOMICAL GRAVITATIONAL CYCLE SUDDEN WATER DISPLACEMENT TOTAL WATER LEVEL INTERACTING BUT DISTINCT COMPONENTS

HAZARD / ACTOR / INSTRUMENT

STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

TSUNAMI → sudden water displacement
TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT

STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle

HAZARD / ACTOR / INSTRUMENT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

- TSUNAMI → sudden water displacement
- TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOTAL WATER LEVEL → interacting but distinct components.

03

STAGE 03 IMD INFORMATION RAIL

IMD INFORMATION RAIL 1 OBSERVE 2 CLASSIFY FORECAST TRACK AND INTENSITY 3 ISSUE IMPACT AND HAZARD INFORMATION 4 UPDATE AS UNCERTAINTY CHANGES

HAZARD / RISK / RESILIENCE CONCEPT

- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 ISSUE IMPACT AND HAZARD INFORMATION
- 4 UPDATE AS UNCERTAINTY CHANGES

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOTAL WATER LEVEL → interacting but distinct components.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE IMPACT AND HAZARD INFORMATION; 4 UPDATE AS UNCERTAINTY CHANGES.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATCH / ALERT / WARNING / POST-LANDFALL — BULLETIN STAGE.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAINED MANAGEMENT AND MAINTENANCE?; NO → ASSET EXISTS BUT READINESS UNPROVEN.

06

STAGE 06SHELTER READINESS TEST

SHELTER READINESS TESTSAFE SITE AND ACCESS?WATER SANITATION POWER ACCESSIBILITY?TRAINED MANAGEMENT AND MAINTENANCE?ASSET EXISTS BUT READINESS UNPROVEN

SAFE SITE AND ACCESS?

WATER SANITATION POWER ACCESSIBILITY?

TRAINED MANAGEMENT AND MAINTENANCE?

NO – ASSET EXISTS BUT READINESS UNPROVEN

HAZARD / RISK / RESILIENCE CONCEPT

• SAFE SITE AND ACCESS?

• WATER SANITATION POWER ACCESSIBILITY?

INSTITUTIONAL OR COMMUNITY RESPONSE

• TRAINED MANAGEMENT AND MAINTENANCE?

• NO – ASSET EXISTS BUT READINESS UNPROVEN

CAPACITY / STATUS / EVIDENCE LIMIT

• SAFE SITE AND ACCESS?

• WATER SANITATION POWER ACCESSIBILITY?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAINED MANAGEMENT AND MAINTENANCE?; NO – ASSET EXISTS BUT READINESS UNPROVEN.

07

STAGE 07CRITICAL-SERVICE WEB

CRITICAL-SERVICE WEBSUPPLY CHAINEOC / HOSPITAL / RESCUEPOWER / WATER / TELECOM

EOC / HOSPITAL / RESCUE

POWER / WATER / TELECOM

ROAD / PORT / SUPPLY CHAIN

REDUNDANCY – ASSESS – RESTORE

HAZARD / RISK / RESILIENCE CONCEPT

• EOC / HOSPITAL / RESCUE

• POWER / WATER / TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE

• ROAD / PORT / SUPPLY CHAIN

• REDUNDANCY – ASSESS – RESTORE

CAPACITY / STATUS / EVIDENCE LIMIT

• EOC / HOSPITAL / RESCUE

• POWER / WATER / TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROAD / PORT / SUPPLY CHAIN; REDUNDANCY – ASSESS – RESTORE.

08

STAGE 08HOUSING-TO-ECOSYSTEM PORTFOLIO

HOUSING-TO-ECOSYSTEM PORTFOLIOSITING CODE COMPLIANCE MAINTENANCELIFELINE HARDENING AND REDUNDANCYMANGROVE DUNE WETLAND PROTECTIONLAYERED DEFENCE WITH RESIDUAL RISK

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
HOUSING → siting code compliance maintenance	ECOSYSTEM → mangrove dune wetland protection	HOUSING → siting code compliance maintenance
INFRASTRUCTURE → lifeline hardening and redundancy	RULE → layered defence with residual risk	INFRASTRUCTURE → lifeline hardening and redundancy

HAZARD / ACTOR / INSTRUMENT

• HOUSING → siting code compliance maintenance

• INFRASTRUCTURE → lifeline hardening and redundancy

DECISIVE RISK OR MANDATE DISTINCTION

• ECOSYSTEM → mangrove dune wetland protection

• RULE → layered defence with residual risk

WARNING / IMPLEMENTATION / OUTCOME LIMIT

• HOUSING → siting code compliance maintenance

• INFRASTRUCTURE → lifeline hardening and redundancy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ECOSYSTEM → mangrove dune wetland protection; RULE → layered defence with residual risk.

09

STAGE 09NCRMP STATUS LADDER

NCRMP STATUS LADDERPROJECT APPROVALASSET CREATIONPHASE COMPLETIONLOCAL HAND-OFFOPERATION + MAINTENANCE + DRILLSCURRENT READINESSNEEDS SEPARATE EVIDENCE

PROJECT APPROVAL – ASSET CREATION

PHASE COMPLETION – STATE / LOCAL HAND-OFF

OPERATION + MAINTENANCE + DRILLS

CURRENT READINESS – needs separate evidence

RISK CONDITION / INPUT

• PROJECT APPROVAL – ASSET CREATION

• PHASE COMPLETION – STATE / LOCAL HAND-OFF

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

• OPERATION + MAINTENANCE + DRILLS

• CURRENT READINESS – needs separate evidence

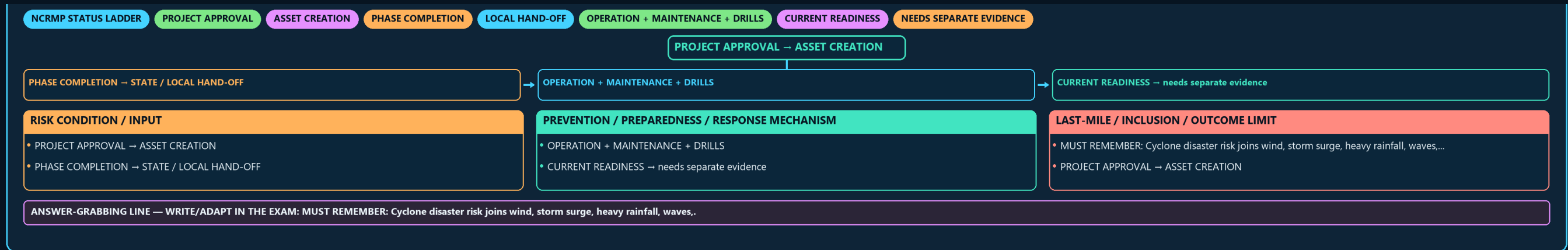
LAST-MILE / INCLUSION / OUTCOME LIMIT

• MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,...

• PROJECT APPROVAL – ASSET CREATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,.

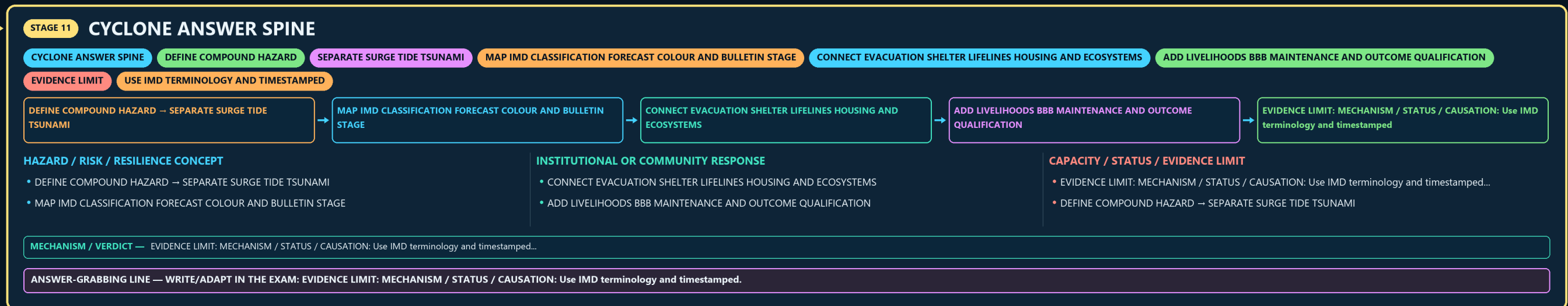
CYCLONES, STORM SURGE AND COASTAL PREPAREDNESS • TILE 3/4 • same master rows 4968-8202 of 10686 • continues below



10



11



E

